

Supplementary Information

Yuxing Hao^{1*}, Kumail Alhamoud¹, Hyewon Jeong¹, Haoran Zhang¹, Grace Yan¹, Isha Puri¹, Philip Torr³, Mike Schaekermann⁶, Saleh Kalantari², Ariel D. Stern^{4,5}, and Marzyeh Ghassemi¹

¹Massachusetts Institute of Technology, Cambridge, MA, USA

²Cornell University, Ithaca, NY, USA

³University of Oxford, Oxford, UK

⁴Hasso Plattner Institute, Potsdam, Germany

⁵University of Potsdam, Potsdam, Germany

⁶Independent Researcher

*Corresponding Author: yuxing@mit.edu

June 22, 2026

1 Dataset Explanation

Dataset	Total QA	Avg Words Per Sentence		Perplexity	
		High	Low/Irr	High	Low/Irr
MMLU (Precision Medicine)	192	16.16 (4.08)	12.8 (4.6)	46.4 (56.3)	58.7 (70.4)
JAMA Clinical Challenge	295	20.76 (4.52)	16.0 (5.4)	51.6 (69.3)	68.2 (92.4)
MedBullets	160	15.73 (3.15)	16.0 (4.3)	46.5 (51.1)	48.3 (65.8)
MedXpertQA	286	17.06 (5.07)	15.6 (4.9)	41.4 (43.8)	52.3 (71.0)
Overall	933	17.81 (4.86)	15.4 (5.1)	48.7 (62.0)	61.0 (82.9)

Table 1: **Comparative Analysis of Physician Trainee–Annotated *MedPair* Dataset Characteristics.** Values in parentheses represent standard deviations.

Massive Multitask Language Understanding (MMLU) is a common benchmark which consists of multiple domains and tasks based on real-world exams [?]. It includes 57 subjects across STEM, the humanities, the social sciences. Here we only focused on medical related questions (precision_medicine), which has 272 multiple-choice medical questions.

The **JAMA Clinical Challenge dataset** includes 1,034 clinical cases sourced from the JAMA Network Clinical Challenge archive. Each entry summarizes a real and diagnostically complex clinical scenario, presented in the form of a question. These challenges feature an extended case vignette followed by a multiple-choice question with four answer options, accompanied by a detailed discussion explaining both the correct and incorrect responses. The questions span a broad spectrum of medical topics [?].

Medbullets consists of 298 United States Medical Licensing Examination (USMLE) Step 2 and Step 3 style questions curated from open-access posts beginning in April 2022. These questions aim to reflect common clinical scenarios encountered in medical education, with difficulty levels comparable to Step 2 and 3 exams. Each item includes a brief case description, five answer choices, and an explanation that clarifies the reasoning behind both correct and incorrect responses. Compared to JAMA, these cases tend to be shorter and potentially less complex [?].

MedXpertQA consists of 2450 questions for text evaluation. It is a highly challenging and comprehensive medical multiple-choice benchmark. MedXpertQA integrates specialty-specific assessments into medical benchmarking and challenging medical exam questions with real-world clinical information into medical multimodal benchmarking [?].

1.1 Evaluation Set-Up

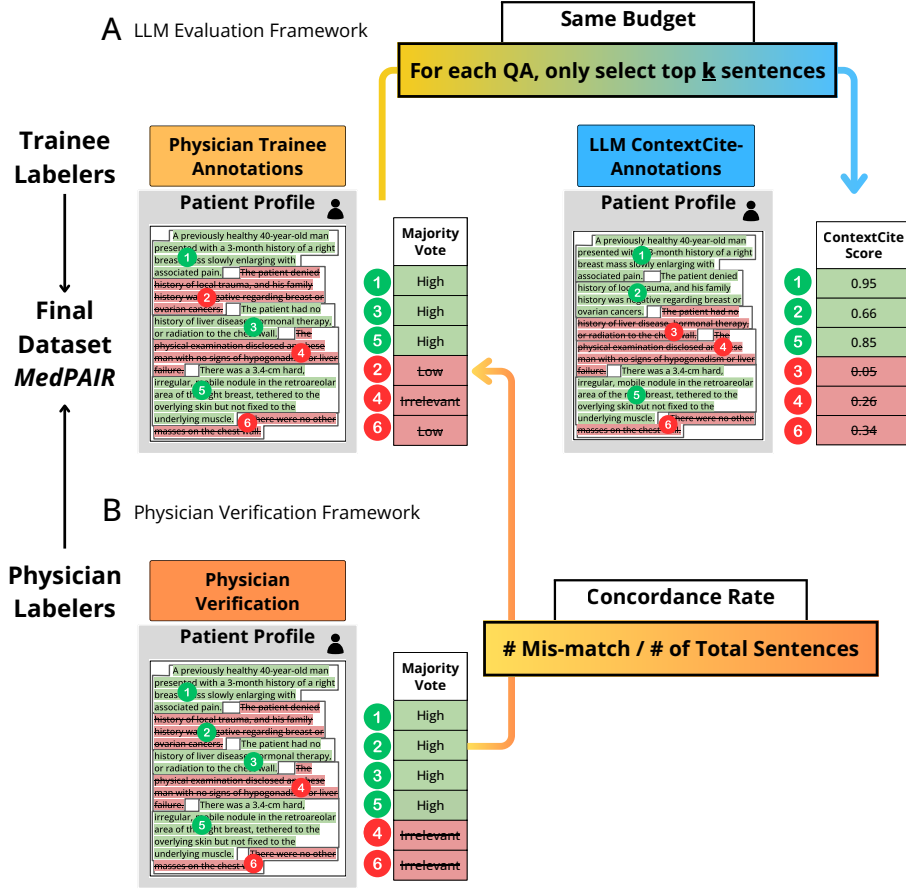


Figure 1: Aligning Physician Trainee Annotations with LLM ContextCite Raw Scores Using an Identical Input Context Budget.

1.2 NLP Analysis

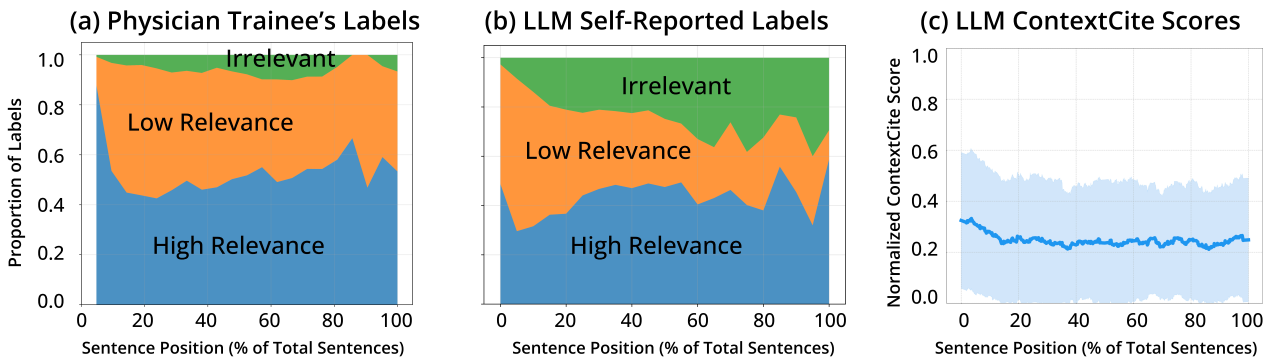


Figure 2: **Sentence Position Analysis.** Plot (a) Distribution of physician trainees majorityvote relevance labels by sentence position. Plot (b) Distribution of GPT-4o self-reported relevance labels by sentence position. Plot (c) ContextCite scores across the context for three open-source models (Qwen-14B, Llama-70B, Qwen-72B).

In order to investigate how sentence relevance shifts according to its position in the clinical vignette, we plotted the labels assigned by physician trainees and those self reported by LLMs (Figure 2 plots (a), (b)) and LLM ContextCite scores (Figure 2 plot (c)). Our objective was to determine whether trainees or the model demonstrate systematic attention to particular segments of the patient profile. All three plots indicate that sentences appearing at the beginning of the text receive the highest relevance ratings. GPT-4o marks slightly fewer sentences as highly relevant and more as irrelevant in the central region compared with physician trainees. In contrast, ContextCite scores decline from approximately 0.33 at the outset to 0.22 by the tenth percentile, then plateau between 0.20 and 0.25 with minimal variance. This flat, low-variance profile diverges sharply from the dynamic patterns of expert and self-reported labels, suggesting that ContextCite does not capture the nuanced, position-dependent relevance judgments.

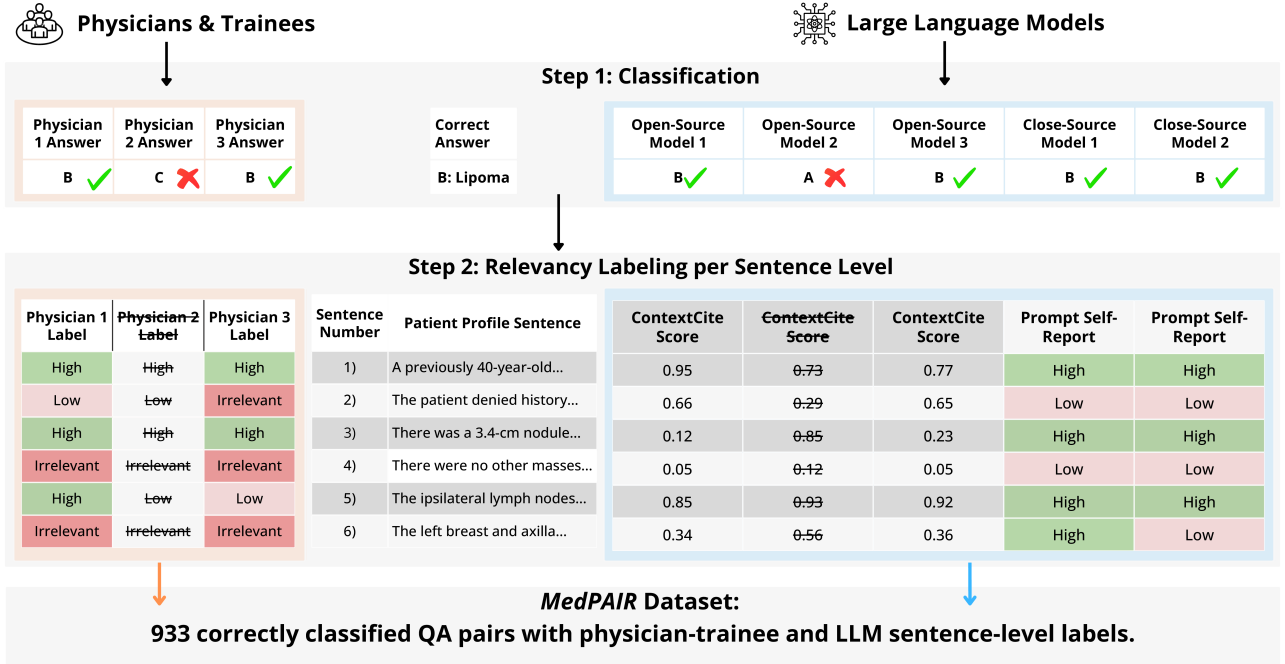


Figure 3: **MedPAIR Dataset Curation.** We curated a 933 annotated QA dataset MedPAIR. Data are structured into patient profile and query components. In the first step, 52 physician trainees and 6 LLMs independently select the most appropriate answer. In the second step, physician trainees annotate the relevance of each sentence within the patient profile, excluding annotations link to incorrect answers. Majority voting is used to produce binary relevance labels for physician trainees. Concurrently, we prompt close-source LLMs (GPT-4o, GPT-5) to follow the physician annotation instructions and employ ContextCite with open-source LLMs (Qwen-14B, Qwen-72B, Llama-70B, MedGemma-27B) to generate both relevance scores and self-reported labels.

1.3 Model Memorizations

We further investigate memorization in open-source models to determine whether MedPAIR data may already be present or have been used during their pre-training. We apply the Min-K%++ membership inference attack [?] to evaluate whether a specific text segment was included in an LLMs training set. For each question, we query the model with whats the probability of this exact text? and compute multiple signal scores based on the token-level log-probabilities.

Since it is an LLM explicitly trained for the medical domain, MedGemma-27B is consistently the most confident on all four medical sources. Among the general-purpose models, Llama-3.1-70B shows slightly higher confidence than the Qwen models. The detailed results are reported in Table 2.

Model	JAMA (n=295)	MedXpert (n=286)	MedBullets (n=160)	MMLU (n=192)
Llama 3.1 70B	-1.572 ± 0.202 [-1.596, -1.549]	-1.622 ± 0.308 [-1.658, -1.586]	-1.469 ± 0.187 [-1.498, -1.439]	-1.589 ± 0.297 [-1.631, -1.546]
Qwen2.5 72B	-1.523 ± 0.217 [-1.548, -1.498]	-1.572 ± 0.333 [-1.610, -1.533]	-1.381 ± 0.195 [-1.411, -1.351]	-1.485 ± 0.306 [-1.529, -1.442]
Qwen2.5 14B	-1.563 ± 0.214 [-1.588, -1.539]	-1.581 ± 0.322 [-1.619, -1.544]	-1.415 ± 0.200 [-1.446, -1.384]	-1.543 ± 0.289 [-1.584, -1.502]
MedGemma 27B	-2.010 ± 0.258 [-2.040, -1.981]	-1.945 ± 0.373 [-1.988, -1.901]	-1.726 ± 0.221 [-1.761, -1.692]	-1.894 ± 0.342 [-1.943, -1.845]

Table 2: **Model Memorization Metrics across Different Datasets.** All resulting scores are negative. Values that are less negative (i.e., closer to 0) indicate that the model considers the text more predictable, which provides a stronger indication that the text appeared in its training data. Values presented as mean $\pm$ standard deviation.

2 Expert Annotation Dataset Interpretation

We instructed each labeler to annotate every sentence as high relevance, low relevance, or irrelevant. Depending on question difficulty and the exclusion of annotations from labelers whose answer classifications proved incorrect, each item received at least two valid annotations. We then assign numeric scores to those labels: high relevance was scored as 1.0, low relevance as 0.5, and irrelevant as 0.0. For each sentence, we calculated the average of those scores. If the average exceeded 0.66, we classified the sentence as high relevance; if it fell between 0.33 and 0.66, we classified it as low relevance; and if it was below 0.33, we classified it as irrelevant.

After the sentence removal, we have the number of sentences in high or low irrelevant Table 3.

Source	#QA	High / QA (mean $\pm$ std)	Low / QA (mean $\pm$ std)	High %	Low %
JAMA	295	8.1 ± 3.2	4.4 ± 2.6	65.0%	35.0%
MedXpert	286	5.4 ± 5.9	2.6 ± 2.1	67.2%	32.8%
MedBullets	160	5.6 ± 1.9	3.8 ± 1.9	59.5%	40.5%
MMLU	192	4.6 ± 2.1	3.0 ± 2.3	60.5%	39.5%
Overall	933	6.1 ± 3.4	3.5 ± 2.4	63.9%	36.1%

Table 3: **Distribution of High and Low or irrelevant Sentences QA Pairs Across Datasets.** Mean and standard deviation represent the average number of high and low/irrelevant annotations per question.

To structure the evaluation, we designed a framework distinguishing between accurate prediction and relevance agreement, summarized in the confusion matrix presented in Table 4. We evaluated the outcomes under both conditions in two ways: (a) *Answer correctness*: did the model get the question right or wrong? and (b) *Relevance agreement*: how well did the model align with the physician trainees annotated relevant components?

We quantified alignment using metrics such as the proportion of the models referenced high/low relevance components and the frequency of referencing irrelevant components, comparing these against the ground-truth annotations. Our goal is to ensure that relevance agreement aligns with both accurate prediction (true positives (TP) in Table 4) and correct relevance, using clinicians’ relevance labels with correct predictions as ground truth. We seek to minimize cases where the model achieves correct predictions but relies on incorrect relevance (false positives (FP) in Table 4).

		Relevance Agreement	
		Yes	No
Accurate Prediction	Yes	TP (Relevance✓, Accurate ✓)	FP (Relevance✗, Accurate ✓)
	No	FN (Relevance✓, Accurate ✗)	TN (Relevance✗, Accurate ✗)

Table 4: **Confusion matrix of prediction accuracy and relevance agreement.** In our MedPAIR benchmark, we evaluated relevance labels from both physician trainee labelers and LLMs.

3 Labeler Instructions & Prompts

We asked each physician trainee or physician labeler to follow this instruction during sentence-level relevance labeling:

You are given a list of sentences from a clinical vignette and a multiple-choice clinical question.

Your task is twofold: (1) Select the most appropriate answer from the given options. (2) Label each sentence as either **[High Relevance]**, **[Low Relevance]**, or **[Irrelevant]**, based on its contribution to answering the question.

DEFINITIONS:

[HIGH RELEVANCE]: Give this label to sentences that directly answer the medical question with specific and essential information. If this part is missing or altered, the answer would be significantly affected.

- A sentence that explicitly states the primary cause or contributing factor (history, demographics, etc.) is considered high relevance.
- If the question is asking about the treatment plan, a sentence that clearly states the specific indication of the proposed treatment plan is considered high relevance.
- If the question is asking about the diagnosis, a sentence that includes diagnostic criteria for the condition is considered high relevance.
- If the question is asking about test results, a sentence that clearly reports the key findings that confirm or support the test outcome is considered high relevance.

[LOW RELEVANCE]: Give this label to sentences that offer background or contextually related or background information that may be helpful but do not directly answer the question.

- A sentence that includes a secondary or potential contributing factor (symptoms, history, etc.) of the main patient condition is considered low relevance.
- A negative history that contradicts or does not support the diagnosis (e.g., no prior epistaxis when the diagnosis is epistaxis) is considered low relevance.
- If the question is asking about the treatment plan, a sentence that includes the intervention or therapy that is not central to the gold standard treatment is considered low relevance.
- If the question is asking about the treatment plan, a sentence that describes the outcome of a previous intervention for the current chief complaint or diagnosis particularly one that was unsuccessful is considered low relevance.
- If the question is asking about the treatment plan, a sentence that does not indicate the treatment itself but instead rules out other conditions that would require different treatments is considered low relevance.
- If the question is asking about the diagnosis, a sentence that includes criteria that could rule out the current diagnosis (that provides differential diagnosis of the patient condition) is considered low relevance.
- If the question is asking about test results, a sentence that reports findings that correlate with or commonly co-occur with the expected result but are not definitive is considered low relevance.

[IRRELEVANT]: Give this label to sentences that do not fall under high or low relevance, or that seem completely unrelated or unhelpful to answering the question. Irrelevant sentences wouldn't affect anyone answering this QA even if this is removed.

- Sentence that adds no additional information on solving question and doesn't help in differentially diagnosing the condition
- General findings, not specific to the diagnosis or management decision.

Focus on identifying the information that directly contributes to answering the question. This task involves only text and does not include any images. If the text refers to figures or mentions from the image, focus only on the information presented in the text. Please consider the following clinical question and answer options when labeling each sentence. Then, label each sentence.

To ensure both physician labelers and the LLM received identical instructions, we used the same prompt when eliciting self-reported sentence-level relevance annotations.

We also asked LLMs to output the answer while compiling the ContextCite score for each sentence.

You are a clinical reasoning assistant. You will receive a patient case summary and a multiple-choice question. Read the question and state your answer.
Patient Context: [patient profile text]
Question and Options: [question and options]
Please select the single most appropriate answer. Respond only in the following format:
Answer: [LETTER]

Answer all the questions (1/582)
View Instructions
Exit labeling

Step 1: Read excerpt

1. A 10-year-old girl is admitted to the medical floor for a respiratory infection.
2. The patient lives in a foster home and has been admitted many times.
3. Since birth, the patient has had repeated episodes of pain/pressure over her frontal sinuses and a chronic cough that produces mucus.
4. She was recently treated with amoxicillin for an infection.
5. The patient is in the 25th percentile for height and weight which has been constant since birth.
6. Her guardians state that the patient has normal bowel movements and has been gaining weight appropriately.
7. The patient has a history of tricuspid stenosis.
8. She also recently had magnetic resonance imaging (MRI) of her chest which demonstrated dilation of her airways.
9. Her temperature is 99.5°F (37.5°C), blood pressure is 90/58 mmHg, pulse is 120/min, respirations are 18/min, and oxygen saturation is 94% on room air.
10. Physical exam is notable for bruises along the patient's shins which the guardians state are from playing soccer.
11. The rest of the exam is deferred because the patient starts crying.

Step 2: Answer QA Details

Which of the following findings is associated with this patient's most likely underlying diagnosis?

Case #25263864

1. Answer QA Question

Select an answer

2. Is sentence 1 clinically relevant?

not applicable

3. Is sentence 2 clinically relevant?

not applicable

4. Is sentence 3 clinically relevant?

not applicable

5. Is sentence 4 clinically relevant?

not applicable

6. Is sentence 5 clinically relevant?

not applicable

7. Is sentence 6 clinically relevant?

Submit

Add comment

Flag case

Figure 4: **Centaur Labs Labeling Interface.** The physician labelers first answer the classification question, then provide "high relevance", "low relevance", and "not relevant" labels to each sentence. Physician labelers can also add comments or flag cases during labeling.

For Supplementary Table 6, we calculated trainee and physician accuracy by randomly selecting one response per unique question ID to ensure each question was counted only once to avoid duplicate-response bias.

Data Source	Qwen-72B		Llama-70B		Qwen-14B		MedGemma	GPT-4o	GPT-5	Physicians
	CC	SR	CC	SR	CC	SR	SR	SR	SR	SR
MMLU	26.9 (22.2)	72.6 (21.2)	70.7 (24.7)	57.4 (21.3)	56.9 (19.6)	53.6 (23.9)	66.1 (21.9)	50.5 (29.4)	60.3 (17.0)	77.1 (18.2)
JAMA	45.5 (23.9)	69.0 (16.7)	62.1 (22.6)	57.5 (16.1)	59.1 (20.1)	55.9 (18.3)	63.1 (17.4)	45.2 (29.3)	35.7 (18.2)	69.2 (16.8)
MedBullets	49.8 (24.8)	68.9 (18.7)	66.6 (23.6)	52.1 (18.7)	53.9 (17.8)	47.7 (20.6)	61.3 (19.2)	45.2 (28.4)	50.2 (11.3)	72.1 (16.0)
MedXpertQA	51.8 (30.3)	69.3 (21.3)	69.3 (27.2)	53.2 (22.2)	51.9 (19.9)	50.4 (24.0)	63.3 (24.0)	52.1 (35.4)	64.3 (12.2)	74.1 (19.4)
Overall	44.9 (26.7)	69.6 (19.0)	65.9 (24.5)	55.6 (19.1)	56.2 (19.8)	52.9 (21.2)	63.3 (20.2)	47.7 (30.9)	48.7 (20.1)	72.0 (17.7)

Table 5: **Relevance Label Concordance (%)** with Physician Trainee Labels. Standard deviations presented in parentheses. "CC" denotes ContextCite score; "SR" denotes Self-Reported labels.

Answer all the questions (1/502)

View Instructions
Exit labeling

Step 1: Read excerpt

1. A 10-year-old girl is admitted to the medical floor for a respiratory infection.
2. The patient lives in a foster home and has been admitted many times.
3. Since birth, the patient has had repeated episodes of pain/pressure over her frontal sinuses and a chronic cough that produces mucus.
4. She was recently treated with amoxicillin for an infection.
5. The patient is in the 25th percentile for height and weight which has been constant since birth.
6. Her guardians state that the patient has normal bowel movements and has been gaining weight appropriately.
7. The patient has a history of tricuspid stenosis.
8. She also recently had magnetic resonance imaging (MRI) of her chest which demonstrated dilation of her airways.
9. Her temperature is 99.5°F (37.5°C), blood pressure is 90/58 mmHg, pulse is 120/min, respirations are 18/min, and oxygen saturation is 94% on room air.
10. Physical exam is notable for bruises along the patient's shins which the guardians state are from playing soccer.
11. The rest of the exam is deferred because the patient starts crying.

Step 2: Answer QA Details

Which of the following findings is associated with this patient's most likely underlying diagnosis?

Case #25263864

1. Answer QA Question

Select an answer

2. Is sentence 1 clinically relevant?

not applicable

3. Is sentence 2 clinically relevant?

not applicable

4. Is sentence 3 clinically relevant?

not applicable

5. Is sentence 4 clinically relevant?

not applicable

6. Is sentence 5 clinically relevant?

not applicable

7. Is sentence 6 clinically relevant?

Submit

Add comment

Flag case

Figure 5: Centaur Labs Labeling Interface. The physician labelers first answer the classification question, then provide "high relevance", "low relevance", and "not relevant" labels to each sentence. Physician labelers can also add comments or flag cases during labeling.

Base Model	Condition	MMLU	Jama	MedXpert	Medbullets	General
<i>Trainee Removed (Low+Irr Labeled by Trainees) – Detailed Accuracies – 933 QAs</i>						
GPT5	Trainee Removed	97.9	89.2	74.1	95.0	87.4
GPT4o	Trainee Removed	96.4	82.0	43.4	86.9	74.0
Llama-70B	Trainee Removed	91.7	73.9	34.3	78.1	66.1
Qwen-72B	Trainee Removed	92.2	71.5	30.8	75.6	64.0
Qwen-14B	Trainee Removed	84.9	68.5	24.8	60.6	57.1
MedGemma-27B	Trainee Removed	45.3	20.0	11.9	31.2	24.7
<i>Physician-Irrelevant (Low+Irr Sentences Only) – Detailed Accuracies – 933 QAs</i>						
GPT5	Physician-Irrelevant	73.4	61.7	43.0	55.6	57.3
GPT4o	Physician-Irrelevant	96.4	82.0	43.4	86.9	74.0
Llama-70B	Physician-Irrelevant	54.7	48.5	23.8	41.2	40.9
Qwen-72B	Physician-Irrelevant	55.7	46.4	19.6	38.8	38.8
Qwen-14B	Physician-Irrelevant	51.6	46.8	20.6	35.6	37.8
MedGemma-27B	Physician-Irrelevant	19.3	16.3	12.6	21.9	16.7
<i>Physician-Random (Random Sentence Subset Baseline) – Detailed Accuracies – 933 QAs</i>						
GPT5	Physician-Random	93.8	84.1	59.4	81.9	78.1
GPT4o	Physician-Random	56.8	59.3	46.2	58.8	54.7
Llama-70B	Physician-Random	81.8	65.8	29.0	65.0	57.7
Qwen-72B	Physician-Random	83.9	61.0	26.2	63.7	55.5
Qwen-14B	Physician-Random	79.2	63.4	24.5	54.4	53.2
MedGemma-27B	Physician-Random	37.0	19.3	12.9	28.7	22.6
<i>Qwen-14B Removed (CC) – Detailed Accuracies – 933 QAs</i>						
GPT4o	Qwen-14B Removed	95.3	76.8	40.3	79.7	71.1
GPT5		97.7	87.7	67.6	91.4	84.7
Llama-70B		91.7	73.9	34.3	78.1	66.1
Qwen-72B		92.2	71.5	30.8	75.6	64.0
Qwen-14B		84.9	68.5	24.8	60.6	57.1

Base Model	Condition	MMLU	Jama	MedXpert	Medbullets	General
MedGemma-27B		45.3	20.0	11.9	31.2	24.7
Qwen-72B Removed (CC) – Detailed Accuracies – 933 QAs						
GPT4o	Qwen-72B Removed	95.9	75.9	41.5	82.5	71.5
GPT5		96.6	83.7	64.6	85.4	81.1
Llama-70B		91.7	73.9	34.3	78.1	66.1
Qwen-72B		92.2	71.5	30.8	75.6	64.0
Qwen-14B		84.9	68.5	24.8	60.6	57.1
MedGemma-27B		45.3	20.0	11.9	31.2	24.7
Llama-70B Removed (CC) – Detailed Accuracies – 933 QAs						
GPT4o	Llama-70B Removed	93.2	69.2	38.8	74.5	66.2
GPT5		96.6	83.2	60.5	84.3	79.8
Llama-70B		91.7	73.9	34.3	78.1	66.1
Qwen-72B		92.2	71.5	30.8	75.6	64.0
Qwen-14B		84.9	68.5	24.8	60.6	57.1
MedGemma-27B		45.3	20.0	11.9	31.2	24.7
Qwen-14B Removed (SR) – Detailed Accuracies – 933 QAs						
GPT-4o	Qwen-14B Removed	95.3	81.4	42.0	84.4	72.7
GPT-5		88.0	87.5	68.5	90.6	82.3
Llama-70B		75.0	49.8	20.2	59.0	47.5
MedGemma-27B		40.8	14.9	4.0	19.1	17.0
Qwen-14B		81.8	59.7	24.2	56.5	52.8
Qwen-72B		79.7	64.1	27.4	58.4	55.1
Qwen-72B Removed (SR) – Detailed Accuracies – 933 QAs						
GPT-4o	Qwen-72B Removed	95.8	79.3	42.3	86.2	72.6
GPT-5		96.9	85.1	68.9	90.0	83.4
Llama-70B		91.7	71.9	32.5	72.5	64.0
MedGemma-27B		93.1	60.0	30.8	65.2	59.5
Qwen-14B		85.4	64.1	25.2	62.5	56.3
Qwen-72B		82.3	69.2	27.3	62.5	57.9
Llama-70B Removed (SR) – Data Not Yet Available						
GPT-4o Removed – Detailed Accuracies – 933 QAs						
GPT4o	GPT-4o Removed	95.9	75.9	41.5	82.5	71.5
GPT5		96.4	78.7	41.2	84.5	73.1
Llama-70B		91.7	73.9	34.3	78.1	66.1
Qwen-72B		92.2	71.5	30.8	75.6	64.0
Qwen-14B		84.9	68.5	24.8	60.6	57.1
MedGemma-27B		45.3	20.0	11.9	31.2	24.7
GPT-5 Removed – Detailed Accuracies – 933 QAs						
GPT4o	GPT-5 Removed	95.9	77.8	40.3	80.6	71.7
GPT5		97.9	86.6	67.0	91.3	84.2
Llama-70B		91.7	73.9	34.3	78.1	66.1
Qwen-72B		92.2	71.5	30.8	75.6	64.0
Qwen-14B		84.9	68.5	24.8	60.6	57.1
MedGemma-27B		45.3	20.0	11.9	31.2	24.7
MedGemma-27B Removed – Detailed Accuracies – 933 QAs						
GPT4o		95.3	74.9	40.1	78.4	69.9

Base Model	Condition	MMLU	Jama	MedXpert	Medbullets	General
GPT5		96.9	86.8	69.2	87.9	84.2
Llama-70B		91.7	73.9	34.3	78.1	66.1
Qwen-72B		92.2	71.5	30.8	75.6	64.0
Qwen-14B		84.9	68.5	24.8	60.6	57.1
MedGemma-27B		45.3	20.0	11.9	31.2	24.7

Table 6: **Comparison of accuracy (%) across datasets** before and after removing sentences that physician trainees or LLMs labeled as low relevance or irrelevant.

Dataset	Readability		Top Keywords Frequency	
	High	Low/Irr	High	Low/Irr
MMLU (Precision Medicine)	62.0 (15.3)	67.4 (26.3)	days, emergency, department, shortness leukocyte , urine , previously	unremarkable, currently , smoke , controlled, illicit , bmi , illness, oral, kgm, weighs
JAMA Clinical Challenge	34.5 (15.3)	40.1 (18.3)	foveal , hyperreflective, spots, girl , progressively , hypopyon , cytoplasm , punctate, ventricular , man	swab , travel, procedures , order , allergic , noncontributory, pertinent, digital , empirically, animals
MedBullets	67.0 (12.2)	71.9 (16.4)	extremity , increased , meq/L, developed , flexion, bright, right , lateral , poor, progressively	metformin , sexually, active, clear , medications , uncomplicated, known, cervical , nonfocal , albuterol
MedXpertQA	49.8 (16.0)	54.9 (21.5)	progressive , severe , levels, low , urea , spine , nitrogen , labor, iliac , mmoll	trauma, allergies , appropriately, murmurs , vitamin , beers , personal, resuscitation , ordered, taking
Overall	47.5 (19.9)	52.9 (23.9)	spots, foveal , hyperreflective , progressive , hypopyon , watery , cytoplasm, punctate, particularly, wrist	organomegaly , smoke , walks , comfortable, noncontributory, personal, nonfocal , antihypertensive , weekends , case

Table 7: **Comparison of dataset characteristics focusing on Readability and Top Keywords Frequency.** Values in parentheses represent standard deviations. The readability is calculated through Flesch Reading Ease score, which typically ranges from 0 to 100, where a higher score indicates that the text is easier to read, and a lower score suggests the text is more difficult. We highlighted each clinical term using different colors based on the type of information it conveys: **symptoms**, **severity**, **description on findings**, **demographics / history**, **medicine**, **medical test**, **anatomical structure/term**, negative findings or suggestive of good patient status, **timeline**, **comparative**.

Model	Avg Pairwise Jaccard to Others (%)	Jaccard vs Human (%)
Qwen-72B (SR)	57.70	63.48
MedGemma-27B (SR)	56.81	61.01
Human	54.89	100.00
GPT-5 (SR)	53.99	61.15
Llama-70B (SR)	53.89	54.60
Qwen-14B (CC)	47.18	57.84
Llama-70B (CC)	46.27	55.73
Qwen-72B (CC)	46.23	55.80
Qwen-14B (SR)	45.59	43.99
GPT-4o (SR)	40.29	40.45

Table 8: **Model-level concordance ranking on the shared 623-origin subset.** We report each model’s average pairwise Jaccard agreement with all other labelers and Jaccard agreement with human trainee majority-vote high-sentence sets. SR = self-reported labels; CC = ContextCite top- K labels.

4 Sample QA Case Study

Example of QA Data (from MedXpert QA ID0071)

Patient Profile:

1. A 29-year-old female presents with low back pain of five days duration.
2. Her new job involves walking several miles daily across a large facility.
3. The pain is localized without radiation; no traumatic history.
4. Medications: only oral contraceptives.

Question: What is the most likely diagnosis?

Options:

- A. bilateral sacral extension
- B. bilateral sacral flexion
- C. sacral base posterior
- D. right-on-right sacral torsion
- E. sacral base anterior
- F. right-on-left sacral torsion
- G. unilateral sacral flexion on the right
- H. left-on-left sacral torsion
- I. left-on-right sacral torsion
- J. unilateral sacral extension on the left

Correct Answer: (D) right-on-right sacral torsion.

Sentence-level relevance labels

SR = Self-Reported labels; CC = ContextCite labels; unselected / no-label entries are shown as Low/Irr.

Sentence #	Trainee Majority Vote	GPT-4o SR	GPT-5 SR	Qwen-14B SR	Qwen-72B SR	Llama-70B SR	MedGemma-27B SR	Qwen-14B CC	Qwen-72B CC	Llama-70B CC
1	High	Low	Low	Low	Low	High	Low	High	High	High
2	Low/Irr	Low	High	Low	Low	Low/Irr	Low	High	Low/Irr	Low/Irr
3	Low/Irr	High	Low	Low	High	Low/Irr	Low	High	Low/Irr	Low/Irr
4	Low/Irr	Low/Irr	Irr	Irr	Irr	Low/Irr	Irr	Low/Irr	Low/Irr	Low/Irr

In this case study, we observe notable disagreement in informativeness assessments across sentence 2 and sentence 3 among physicians, GPT-4o, and LLaMA-70B ContextCite. Sentence 2 (Her new job involves walking several miles daily across a large facility) was labeled as *Low* by both physicians and GPT-4o, yet *High* by LLaMA-70B ContextCite. This sentence describes the patients lifestyle, specifically her physical activity level related to her job. This information is of limited relevance to sacral torsion. While one cannot entirely rule out its contributionsince prolonged walking with an asymmetric posture could potentially predispose a patient to sacral dysfunctionit is not a direct cause or a diagnostically decisive factor. As such, it offers minimal value in determining the correct answer to this question. LLaMA-70B may have overemphasized contextual lifestyle clues, interpreting the exertion from walking as highly indicative of a mechanical sacral dysfunction, whereas clinicians likely viewed it as a nonspecific background factor without clear diagnostic utility.

The sentence 3 (The pain is localized without radiation; no traumatic history) received a Low label from physicians but High from GPT-4o and LLaMA-70B. This discrepancy may reflect differing heuristics: while clinicians might not prioritize localization and absence of trauma due to their non-specificity or commonality in musculoskeletal complaints, models may have heuristically linked "localized pain without radiation"

to mechanical causes, interpreting it as informative. These examples illustrate how LLMs may misattribute diagnostic weight to surface-level patterns.

5 Sample QA Case Study

Example of QA Data (from JAMA QA ID1049)

Patient Profile:

1. A 10-year-old girl developed an annular skin lesion on her anterior neck.
2. The lesion, initially attributed to a bug bite by her mother, enlarged and developed central clearing.
3. The patient lived in North Carolina with no recent travel or known tick bite.
4. She had no symptoms other than mild local itching.
5. After several weeks, the patient's mother applied over-the-counter topical antifungal cream and the lesion faded.
6. After 10 days of antifungal therapy, the patient was evaluated by her pediatrician.
7. Examination results were normal except for a 2.0-cm annular lesion with central clearing and peripheral flaking (Figure).
8. The pediatrician ordered serologic testing for Lyme disease, prescribed 14 days of doxycycline based on the results (Table), and referred the patient to an infectious disease specialist for evaluation.
9. Complete 14 days of antibiotic treatment for Lyme disease.
10. No further testing is indicated.
11. Complete 14 days of antibiotic treatment for Lyme disease.
12. Repeat Lyme disease serology in 1 month to evaluate response.

Question: What would you do next?

Options:

- A. Complete 14 days of antibiotic treatment for Lyme disease. No further testing is indicated.
- B. Complete 14 days of antibiotic treatment for Lyme disease. Repeat Lyme disease serology in 1 month to evaluate response.
- C. Obtain skin biopsy for *Borrelia burgdorferi* PCR.
- D. Discontinue doxycycline and restart topical antifungal cream.

Correct Answer: (D) Discontinue doxycycline and restart topical antifungal cream.

Sentence-level relevance labels

SR = Self-Reported labels; CC = ContextCite labels; unselected / no-label entries are shown as Low/Irr.

Sentence #	Trainee Majority Vote	GPT-4o SR	GPT-5 SR	Qwen-14B SR	Qwen-72B SR	Llama-70B SR	MedGemma-27B SR	Qwen-14B CC	Qwen-72B CC	Llama-70B CC
1	High	High	High	Low	High	Low	Low	High	Low/Irr	Low/Irr
2	High	High	High	Low	High	Low	High	Low/Irr	High	High
3	Low/Irr	Low	Low	Low	Low	Low	Low	High	Low/Irr	Low/Irr
4	Low/Irr	Low	High	Low	Low	Irr	Low	Low/Irr	Low/Irr	Low/Irr
5	High	Low	High	Low	Low	Irr	Low	Low/Irr	Low/Irr	Low/Irr
6	Low/Irr	Low	Low	Low	Low	Low	Low	Low/Irr	Low/Irr	Low/Irr
7	High	High	High	Low	High	Low	High	Low/Irr	High	High
8	Low/Irr	High	High	High	High	High	High	High	Low/Irr	High
9	Low/Irr	High	Irr	High	High	High	High	High	Low/Irr	High
10	Low/Irr	High	Irr	Irr	Low	Irr	Irr	Low/Irr	Low/Irr	Low/Irr
11	Low/Irr	High	Irr	High	High	High	High	Low/Irr	Low/Irr	High
12	Low/Irr	Irr	Irr	High	High	High	High	High	High	High

Patient Profile:

1. A 41-year-old woman, gravida 3, para 3, presents with heavy menstrual bleeding (HMB), dysmenorrhea, and deep dyspareunia.
2. Menses are regular, occurring once every 25 to 26 days and lasting 8 days, with no intermenstrual bleeding.
3. The patient soaks through one pad per hour for the first 3 days of her cycle, and she rates cramping on her heaviest bleeding days as 9 on a 10-point scale.
4. Her past medical history is significant for 3 prior uncomplicated cesarean births with tubal sterilization during her last birth.
5. Vital signs are within normal limits, and body mass index is 22 kg/m2.
6. Pelvic exam shows a 10-week-sized boggy uterus with smooth contours that is mildly tender to palpation.
7. No discrete masses are palpable, and the adnexa are nontender.
8. A urine pregnancy test is negative, and serum bloodwork shows mild anemia.
9. A pelvic ultrasound is ordered.

Question: Which of the following findings would confirm the most likely diagnosis?

Options:

- A. Multiple hyperechoic foci with posterior shadowing in the myometrium
- B. A focal hypoechoic lesion with irregular borders in the endometrium
- C. Endometrial polypoid structures with central vascularity
- D. Thickened endometrial stripe with irregular vascularity
- E. Symmetrical, homogeneous endometrial thickening
- F. Asymmetrical, irregular endometrial thickening
- G. A large, circular hypoechoic structure within the myometrium with posterior shadowing
- H. Diffuse myometrial calcifications with acoustic shadowing
- I. Symmetrical, homogeneous myometrial thickening
- J. Circular anechoic structures within the myometrium without posterior shadowing

Correct Answer: (J) Circular anechoic structures within the myometrium without posterior shadowing.

Sentence-level relevance labels

SR = Self-Reported labels; CC = ContextCite labels; unselected / no-label entries are shown as Low/Irr.

Sentence #	Trainee Majority Vote	GPT-4o SR	GPT-5 SR	Qwen-14B SR	Qwen-72B SR	Llama-70B SR	MedGemma-27B SR	Qwen-14B CC	Qwen-72B CC	Llama-70B CC
1	High	High	High	Low	High	Low	High	High	High	High
2	Low/Irr	Low	Low	Low	Low	Irr	Low	Low/Irr	Low/Irr	Low/Irr
3	Low/Irr	High	High	Low	High	Low	High	Low/Irr	High	High
4	Low/Irr	High	Low	Irr	Low	Irr	Low	Low/Irr	Low/Irr	Low/Irr
5	High	Low	Irr	Irr	Irr	Irr	Irr	Low/Irr	High	High
6	High	Low	High	Low	High	High	High	High	Low/Irr	Low/Irr
7	Low/Irr	Irr	High	Low	Low	Low	Low	Low/Irr	Low/Irr	Low/Irr
8	Low/Irr	Low/Irr	Low	Irr	Low	Low	Low	Low/Irr	Low/Irr	Low/Irr
9	Low/Irr	Low/Irr	Irr	Low	High	High	High	High	Low/Irr	Low/Irr

5.1 Spurious case: GPT-5 on Medbullets ID0102

The case below is a Spurious Rate instance for GPT-5. In Round 1 the model received the full vignette and selected the correct answer (D) *Transient tachypnea of the newborn*. In Round 2 we removed the sentences that the physician trainee majority vote labelled low relevance or irrelevant, and the model switched to (C) *Neonatal respiratory distress syndrome*. This flip contributes to the Spurious Rate reported for GPT-5 in the main text (Section 2.3, Sankey figure). We identified the case by joining the Round 1 predictions in `gpt5_predictions_Original_Accuracy.csv` with the Round 2 predictions in `gpt5_predictions_on_trainee_removal.csv` and filtering on `Original=answer_corr` and `Trainee Removal_extracted≠answer_corr`. The aggregated per source counts are written to `Figures/sankey/data/sankey_spurious_expert933_results.csv` by `Figures/sankey/notebooks/make_llm_sankey_all_models.py`, which is the source file for the Sankey visualisations.

Example of QA Data (from Medbullets QA ID0102)

Patient Profile:

1. A newborn boy is evaluated 30 minutes after birth.
2. He was born at 39 weeks gestation to a 27-year-old primigravid mother via cesarean section for cervical insufficiency.
3. The pregnancy was complicated by gestational diabetes and the amniotic fluid was clear.
4. Upon delivery, the patient had strong respiratory effort and a strong cry.
5. His Apgar scores at 1 and 5 minutes were 7 and 8, respectively.
6. The patient now is exhibiting increased work of breathing and is progressively more tachypneic.
7. His birth weight is 3,568 g (7 lb 14 oz).
8. His temperature is 99.0°F (37.2°C), blood pressure is 60/44 mmHg, pulse is 146/min, and respirations are 72/min.
9. On physical exam, the patient is grunting with nasal flaring and subcostal retractions.
10. Breath sounds are decreased at the bases bilaterally.
11. The patient has central cyanosis.
12. His chest radiograph can be seen in Figure A.

Question: Which of the following is the most likely etiology of this patient's presentation?

Options:

- A. Meconium aspiration syndrome
- B. Persistent pulmonary hypertension
- C. Neonatal respiratory distress syndrome
- D. Transient tachypnea of the newborn

Correct Answer: (D) Transient tachypnea of the newborn.

GPT-5 Round 1 answer (full context): (D) correct.

GPT-5 Round 2 answer (after removing trainee low/irr sentences): (C) incorrect.

Sentence-level relevance labels

SR = Self-Reported labels; CC = ContextCite labels; unselected / no-label entries are shown as Low/Irr.

Sentence #	Trainee Majority Vote	GPT-4o SR	GPT-5 SR	Qwen-14B SR	Qwen-72B SR	Llama-70B SR	MedGemma-27B SR	Qwen-14B CC	Qwen-72B CC	Llama-70B CC
1	Low/Irr	Low	High	Low	Low	Low	Low	High	Low/Irr	Low/Irr
2	Low/Irr	High	High	Low	Low	Low	Low	Low/Irr	Low/Irr	Low/Irr
3	Low/Irr	Low	High	Low	Low	Low	Low	Low/Irr	Low/Irr	Low/Irr
4	Low/Irr	Low	High	High	Low	High	High	Low/Irr	Low/Irr	Low/Irr
5	Low/Irr	Low	High	Low	Low	Low	High	Low/Irr	Low/Irr	Low/Irr
6	High	High	High	High	High	High	High	High	High	High
7	Low/Irr	Irr	Low	Low	Low	Low	Low	Low/Irr	Low/Irr	Low/Irr
8	Low/Irr	Low	High	High	Low	Low	High	Low/Irr	Low/Irr	Low/Irr
9	High	High	High	High	High	High	High	Low/Irr	High	High
10	High	High	Low	High	High	High	High	Low/Irr	High	High
11	Low/Irr	High	Low	High	High	High	High	High	Low/Irr	Low/Irr
12	Low/Irr	Low/Irr	Low	Low	High	High	High	Low/Irr	Low/Irr	Low/Irr

The flip is informative about the limits of the Spurious Rate metric. Sentence 2 (gestation age and delivery by cesarean section) and sentence 3 (gestational diabetes) are textbook risk factors for transient tachypnea of the newborn and help distinguish it from neonatal respiratory distress syndrome, which typically affects preterm infants with surfactant deficiency. The physician trainee majority vote, however, marked these sentences as low relevance, so they were removed in Round 2. GPT-5 initially weighted them (self-labelled High) and chose the correct answer, then defaulted to the more generic Neonatal respiratory distress syndrome once this context was stripped. The case therefore illustrates that a correct to incorrect transition can reflect the removal of genuinely decisive information that the trainees under-weighted, rather than model reliance on spurious cues. We discuss this confound in the Limitations paragraph of the main text.

5.2 Spurious case: JAMA ID0973

Long patient profile (17 sentences) in which four of six LLMs get Round 1 correct and flip to the same wrong answer in Round 2 once the physician trainee identified low or irrelevant sentences are removed. The physician trainees themselves picked option C in both rounds and never reached the correct answer.

Example of QA Data (from JAMA QA ID0973)

Patient Profile:

1. A 31-year-old man presented with left cervical and left inguinal masses.
2. He reported intermittent itching and night sweats for 2 years.
3. He denied fever, weight loss, shortness of breath, rashes, diarrhea, and neurological symptoms.
4. On a preemployment evaluation, the patient was told he had a high white blood cell count 2 years ago.
5. On examination, there was left cervical and inguinal lymphadenopathy and no other organomegaly.
6. Complete blood cell count and peripheral blood smear showed marked leukocytosis, with a white blood cell count of 22,340/ μ L, an absolute neutrophil count of 5360/ μ L, and 55% eosinophils with an absolute eosinophil count of 12,290/ μ L.
7. Vitamin B12 was markedly elevated at more than 4000 pg/mL.
8. The erythrocyte sedimentation rate was 5 mm/h.
9. Lactate dehydrogenase was 180 U/L, and alkaline phosphatase was 81 U/L.
10. Evaluations for HIV and hepatitis B and C were all negative.
11. Serum creatinine, liver enzymes, and total bilirubin were within normal limits.
12. Bone marrow biopsy showed hypercellular marrow, myeloid hyperplasia, increased eosinophils with some dysplasia, and a blast count of 2%.
13. PET/CT scan showed a left upper cervical lymph node of 2.6 cm and a left inguinal lymph node of 3.1 $\times$ 2.3 cm (SUVmax 5.7).
14. Left inguinal lymph node biopsy showed partial involvement by atypical cells with high proliferation index (Ki-67 >95%) positive for CD3, CD4, CD8, BCL2, and TDT, suggestive of T-cell lymphoblastic lymphoma/leukemia.

15. B, Lymph node biopsy immunohistochemical stain with terminal deoxynucleotidyl transferase.
16. The inset shows interphase fluorescence in situ hybridization for FIP1L1::PDGFRA rearrangement (positive).
17. C, PET/CT 12 weeks after treatment initiation.

Question: What Is Your Diagnosis?

Options:

- A. Kimura disease
- B. Classic Hodgkin lymphoma
- C. T-cell acute lymphoblastic lymphoma/leukemia
- D. Myeloid/lymphoid neoplasms with eosinophilia and tyrosine kinase gene fusions

Correct Answer: (D) Myeloid/lymphoid neoplasms with eosinophilia and tyrosine kinase gene fusions.

Sentence-level relevance labels

SR = Self-Reported labels; CC = ContextCite labels; unselected / no-label entries are shown as Low/Irr.

Sentence #	Trainee Majority Vote	GPT-4o SR	GPT-5 SR	Qwen-14B SR	Qwen-72B SR	Llama-70B SR	MedGemma-27B SR	Qwen-14B CC	Qwen-72B CC	Llama-70B CC
1	High	Low	Low	Low	High	High	High	High	Low/Irr	Low/Irr
2	High	Low	Low	Low	High	High	Low	Low/Irr	Low/Irr	Low/Irr
3	Low/Irr	Low	Low	Low	Low	Low/Irr	Low	Low/Irr	Low/Irr	Low/Irr
4	Low/Irr	Low	Low	Low	Low	Low/Irr	Low	High	Low/Irr	Low/Irr
5	Low/Irr	Low	Low	Low	High	Low/Irr	High	Low/Irr	Low/Irr	Low/Irr
6	Low/Irr	High	High	High	High	Low/Irr	High	High	High	Low/Irr
7	Low/Irr	High	High	Low	Low	Low/Irr	High	High	High	High
8	Low/Irr	High	Irr	Low	Irr	Low/Irr	Low	High	Low/Irr	Low/Irr
9	Low/Irr	Low	Irr	Low	Irr	Low/Irr	Low	Low/Irr	Low/Irr	Low/Irr
10	Low/Irr	High	Irr	Low	Low	Low/Irr	Low	High	Low/Irr	Low/Irr
11	Low/Irr	Irr	Irr	Low	Low	Low/Irr	Low	Low/Irr	Low/Irr	Low/Irr
12	Low/Irr	High	High	Low	High	Low/Irr	Low	High	High	High
13	Low/Irr	Irr	Low	Low	High	Low/Irr	Low	Low/Irr	Low/Irr	Low/Irr
14	High	Low/Irr	High	Low	High	High	Low	Low/Irr	Low/Irr	Low/Irr
15	High	Low/Irr	Low	Low	High	High	Irr	Low/Irr	Low/Irr	Low/Irr
16	Low/Irr	Low/Irr	High	High	High	Low/Irr	High	High	High	High
17	High	Low/Irr	Irr	Irr	Irr	High	Irr	High	Low/Irr	High

Round 1 vs. Round 2 answers

Labeler	Round 1 (full)	Round 2 (trainee removed)	Outcome
GPT-5	D ✓	C ×	flip
GPT-4o	C ×	C ×	both wrong
Qwen-14B	D ✓	C ×	flip
Qwen-72B	D ✓	C ×	flip
Llama-70B	B ×	C ×	both wrong
MedGemma-27B	D ✓	C ×	flip
Trainee maj.	C ×	C ×	both wrong

The flip is driven by the removal of sentence 6 (55% eosinophils), sentence 12 (bone marrow eosinophilia with dysplasia) and sentence 16 (FIP1L1::PDGFRA rearrangement). These are the clues that distinguish option D from the lymph-node biopsy pattern that points to option C. When these sentences are stripped, the remaining text emphasises the CD3/CD4/CD8/TDT-positive biopsy and several models, including the trainees themselves, converge on T-cell ALL. The case is a concrete illustration of the confound discussed in the Limitations: a Round 1 to Round 2 flip can reflect the loss of diagnostic information that the trainees marked as low relevance, rather than model reliance on genuinely spurious content.

5.3 Long-context case: JAMA ID1076

Long patient profile (17 sentences) in the 933-item trainee annotated set. This case has a non-null MedGemma-27B Round 2 answer, and most models remain correct after trainee-labeled low/irrelevant sentence removal.

Example of QA Data (from JAMA QA ID1076)

Patient Profile:

1. An African American woman in her 30s with no medical history presented with a 4-year history of diffuse progressive proliferation of dark papules.
2. The patient noted that the lesions initially began on her thighs and rapidly multiplied throughout her skin.
3. Some of the lesions were pruritic and when scratched could be removed, leaving behind a hyperpigmented macule.
4. The patient described a sunburn several weeks prior to the initial eruption.
5. She was taking no medications and these lesions were unchanged throughout an uncomplicated pregnancy.
6. There was no personal or family history of melanoma or eruptive nevi.
7. On physical examination, she had hundreds of 1- to 2-mm dark brown-black monomorphic papules throughout her trunk, extremities, palms, face, tongue, and sclera (Figure, A).
8. No lymphadenopathy was present.
9. Findings from laboratory tests, including a complete blood cell count, comprehensive metabolic panel, and computed tomography of the chest, were unremarkable.
10. A punch biopsy was performed for histopathologic examination (Figure, B-D).
11. A, Photograph of left thigh shows diffuse 1- to 2-mm dark brown-black macules and papules.
12. B and C, Punch biopsy of right arm shows a circumscribed proliferation of pigmented epithelioid melanocytic nests, located primarily at tips of rete ridges.
13. Moderate cytologic atypia is present.
14. Superficial papillary dermis shows prominent pigment incontinence.
15. No mitotic figures are seen.
16. Kamino bodies were not visualized.
17. D, Melan-A immunostain shows low pagetoid migration of melanocytes with a lack of melanocytic confluence at the dermal-epidermal junction.

Question: What Is Your Diagnosis?

Options:

- A. Superficial spreading melanoma
- B. Eruptive disseminated Spitz nevi
- C. Dermatitis papulosa nigra
- D. Metastatic melanoma

Correct Answer: (B) Eruptive disseminated Spitz nevi.

Sentence-level relevance labels

SR = Self-Reported labels; CC = ContextCite labels; unselected / no-label entries are shown as Low/Irr.

Sentence #	Trainee Majority Vote	GPT-4o SR	GPT-5 SR	Qwen-14B SR	Qwen-72B SR	Llama-70B SR	MedGemma-27B SR	Qwen-14B CC	Qwen-72B CC	Llama-70B CC
1	High	High	High	High	High	Low	High	High	Low/Irr	Low/Irr
2	High	High	High	High	High	Low	High	High	High	High
3	Low/Irr	Low	Irr	Low	High	High	Low	High	Low/Irr	Low/Irr
4	Low/Irr	Low	High	Low	Low	Low	Low	Low/Irr	High	High
5	Low/Irr	High	Low	Low	Low	Irr	Low	Low/Irr	Low/Irr	Low/Irr
6	Low/Irr	High	Low	Low	High	Low	Low	High	Low/Irr	Low/Irr
7	High	Low	High	High	High	High	High	Low/Irr	High	High
8	Low/Irr	Low	High	Low	Low	Low	Low	Low/Irr	High	Low/Irr
9	Low/Irr	Low	Low	Low	Low	Low	Low	Low/Irr	High	High
10	Low/Irr	High	Irr	High	High	High	High	Low/Irr	Low/Irr	High
11	High	High	High	Low	High	Irr	Low	High	Low/Irr	High
12	High	Low	High	High	High	High	High	High	High	High
13	High	Low	High	Low	High	High	High	High	High	High
14	High	High	Low	Low	High	High	Low	High	Low/Irr	Low/Irr
15	High	Low	High	Low	High	High	High	Low/Irr	Low/Irr	Low/Irr
16	High	High	Low	Low	High	High	Low	Low/Irr	High	High
17	High	Low/Irr	High	Low	High	High	High	Low/Irr	High	High

Round 1 vs. Round 2 answers

Labeler	Round 1 (full)	Round 2 (trainee removed)	Outcome
GPT-5	B ✓	B ✓	both correct
GPT-4o	B ✓	B ✓	both correct
Qwen-14B	B ✓	B ✓	both correct
Qwen-72B	B ✓	B ✓	both correct
Llama-70B	D ×	B ✓	recover
MedGemma-27B	B ✓	B ✓	both correct
Trainee maj.	B ✓	B ✓	both correct

This long-context example satisfies the selection criterion requested for Section 5.3: it is part of the 933 trainee-labeled cohort and MedGemma-27B has a non-null Round 2 answer. Unlike the previous ID0876 example, removing trainee-labeled low/irrelevant content does not induce a broad collapse in model accuracy here; instead, predictions remain stable for GPT-5, GPT-4o, Qwen-14B, Qwen-72B, and MedGemma-27B, while Llama-70B improves from an incorrect Round 1 answer to the correct Round 2 answer.

5.4 Spurious case: JAMA ID1266

Long patient profile (16 sentences) that contains the most instructive contrast of the three cases. GPT-5 flips from correct to wrong, Qwen-14B recovers from wrong to correct, and the physician trainees themselves improve from Round 1 wrong to Round 2 correct once their own low relevance sentences are removed.

Example of QA Data (from JAMA QA ID1266)

Patient Profile:

1. A 55-year-old white woman presented with 1 week of decreased visual acuity and ache on abduction in the right eye.
2. She had decreased right-sided hearing and tinnitus, a moderate right-sided frontal headache, and an unsteady gait.
3. She did not have diplopia, facial numbness, or paresthesias.
4. She had a history of stage IV breast cancer treated with letrozole after a mastectomy 5 years prior.
5. On initial examination, best-corrected visual acuity was 20/50 OD and 20/25 OS, with dyschromatopsia, relative afferent pupillary defect, and visual-field defects on the right.
6. Her extraocular movements were full.
7. A dilated fundus examination demonstrated a pale right optic nerve.
8. Brain MRI was reported normal, but on review there was thickening and enhancement of both optic nerves (right > left), cranial nerve III on the right, cranial nerve V bilaterally, the pituitary infundibulum, the hypothalamus, and adjacent cavernous sinuses.

9. Leptomeningeal thickening over the convexities of the anterior frontal lobes was also evident.
10. CSF contained a small population of polyclonal CD19+ B cells and CD4+ and CD8+ T cells.
11. Extensive labs for HIV, ANA, anti-dsDNA, ANCA, SPEP, ESR, CRP, CBC, antiganglioside and anti-MAG antibodies, and serum ACE were normal.
12. Whole-body PET/CT and thoracic CT revealed no abnormalities.
13. Clinically, she initially responded to corticosteroids.
14. Over the ensuing 5 years she had multiple similar presentations that also included facial pain and paresthesias.
15. She developed a chronic small-fiber peripheral neuropathy and was treated with mycophenolate mofetil, cyclophosphamide, and IVIG at various points.
16. She remained clinically stable for 4 years and returned with similar symptoms.

Question: What Would You Do Next?

Options:

- A. Administer high-dose corticosteroids
- B. Repeat the magnetic resonance imaging of the brain
- C. Repeat the whole-body positron emission tomography scan
- D. Repeat the testing for angiotensin-converting enzyme in serum and cerebrospinal fluid

Correct Answer: (C) Repeat the whole-body positron emission tomography scan.

Sentence-level relevance labels

SR = Self-Reported labels; CC = ContextCite labels; unselected / no-label entries are shown as Low/Irr.

Sentence #	Trainee Majority Vote	GPT-4o SR	GPT-5 SR	Qwen-14B SR	Qwen-72B SR	Llama-70B SR	MedGemma-27B SR	Qwen-14B CC	Qwen-72B CC	Llama-70B CC
1	High	High	Low	Low	High	High	High	High	Low/Irr	Low/Irr
2	High	High	Low	Low	High	High	High	Low/Irr	High	High
3	Low/Irr	Low	Low	Low	Low	Low	Low	High	High	High
4	High	Low	High	Low	Low	Low	Low	Low/Irr	High	High
5	High	High	Low	Low	High	High	High	High	High	High
6	Low/Irr	Low	Low	Low	Low	Low	Low	High	High	High
7	High	High	Low	High	High	High	High	High	Low/Irr	High
8	High	High	High	High	High	High	High	High	Low/Irr	Low/Irr
9	High	High	High	High	High	High	High	Low/Irr	Low/Irr	Low/Irr
10	High	Low	High	Low	High	Low	Low	High	Low/Irr	Low/Irr
11	Low/Irr	High	High	Low	High	Low	Low	High	High	High
12	Low/Irr	Low	High	Low	Low	Low	Low	Low/Irr	Low/Irr	Low/Irr
13	High	High	High	High	High	High	High	High	Low/Irr	High
14	High	Low	High	Low	Low	Low	Low	High	High	High
15	High	Low	Low	Low	Low	Low	Low	High	High	High
16	High	High	High	High	High	Low	Low	High	Low/Irr	High

Round 1 vs. Round 2 answers

Labeler	Round 1 (full)	Round 2 (trainee removed)	Outcome
GPT-5	C ✓	B ×	flip
GPT-4o	B ×	B ×	both wrong
Qwen-14B	B ×	C ✓	recovery
Qwen-72B	B ×	B ×	both wrong
Llama-70B	A ×	A ×	both wrong
MedGemma-27B	A ×	n/a	n/a
Trainee maj.	B ×	C ✓	recovery

The two removed sentences here are sentence 11 (extensive negative autoimmune panel including serum ACE) and sentence 12 (prior whole-body PET/CT normal). Together they rule out options A and D and make option C (repeat PET/CT after a multi-year interval) the clinically indicated next step. GPT-5 had reasoned in this

direction with the full context, then moved to option B (repeat MRI) once these two sentences were removed, because the remaining MRI findings became the salient anchor. In contrast, the physician trainees and Qwen-14B improved in Round 2: with the same sentences removed their attention shifted toward the PET-based reasoning and they converged on the correct answer. The case illustrates that the same input manipulation can produce opposite directional effects across labellers, and that the Spurious Rate is therefore better read as an item level probe than a global ranking of reasoning quality.

6 Qualitative Survey Analysis

6.1 Pre-Study Survey

Please provide the following demographic and background information:

- **Age:** [0-120]
- **Gender:** [Female/Male/Other]
- **Medical School Year:** [M1-M4]
- **USMLE Step 1 Status:** [Yes/No] Has the student passed USMLE Step 1?
- **Previous Experience:** [0-50] How many previous studies/tasks have they done with Centaur Labs?
- **Medical School:** [Short Answer] Which medical school do they attend?
- **Familiarity with Clinical Challenges:** [1: Not familiar, 2: Some familiarity, 3: High familiarity] Are they familiar with any of the following clinical challenges? JAMA Clinical Challenge, NEJM Image Challenge, NEJM Resident 360
 - **Follow-up:** [1. not at all; 2. occasionally; 3. every one that is released] If so, how regularly do they follow these challenges?
- **Familiarity with MedBullets:** [1: Not familiar, 2: Some familiarity, 3: High familiarity] Are they familiar with the MedBullets website?
 - **Follow-up:** [1. not at all; 2. occasionally; 3. every one that is released] If so, how regularly do they follow the clinical challenges released there?
- **LLM Experience:** [1-5] Which of the following best describes your experience with large language models (LLMs) in clinical or healthcare-related contexts? (Select the option that most closely matches your experience.)
 1. I have a technical understanding of how LLMs work (e.g., architecture, training, fine-tuning).
 2. I have used LLMs in clinical workflows or research (e.g., for documentation, decision support, or data analysis).
 3. I have interacted with LLMs for general or clinical questions (e.g., using ChatGPT to look up clinical facts or guidelines).
 4. I am aware of LLMs but have not used them in any clinical or technical setting.
 5. I am not familiar with LLMs.
- **LLM Readiness Opinion:** [1-5] Which of the following best reflects your view on the readiness of LLMs for clinical deployment? (Select the option that most closely matches your opinion.)
 1. Not ready at all LLMs are unreliable and unsafe for any clinical use.
 2. Ready for limited, low-risk tasks e.g., documentation assistance, summarization.
 3. Ready for clinician-in-the-loop tasks e.g., decision support with human oversight.
 4. Ready for autonomous use in select areas e.g., triage or patient education.
 5. Fully ready for broad clinical deployment with minimal oversight.

6.2 Post-Study Survey

Please answer the following questions about the task you just completed:

- **Of the questions you have just labeled, what percentage of them have you seen verbatim beforehand (e.g. in your classes, or when preparing for USMLE)?** [Short answer: 0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100] Of the questions you have just labeled, what percentage of them have you seen verbatim beforehand (e.g., in your classes, or when preparing for USMLE)?
- **Of the questions you have just labeled, what do you think your accuracy (percentage) was in answering the multiple choice questions?** [Short answer: 0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100] Of the questions you have just labeled, what do you think your accuracy (percentage) was in answering the multiple choice questions?
- **Suppose we were to give the same clinical questions to the current best LLM (e.g. GPT o3, Claude 3.7 Sonnet), what do you think its accuracy would be on the multiple choice questions?** [Short answer: 0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100] Suppose we were to give the same clinical questions to the current best LLM (e.g., GPT o3, Claude 3.7 Sonnet), what do you think its accuracy would be on the multiple choice questions?
- **How confident were you, on average, in your answers for the sentence relevance scores?** [1: Not confident, 2: Slightly confident, 3: Moderately confident, 4: Very confident, 5: Extremely confident] How confident were you, on average, in your answers for the sentence relevance scores?
- **Please rate the degree of ambiguity you perceive in the true relevance labels on a scale from 1 to 5.** [1: Extreme ambiguity, 2: High ambiguity, 3: Moderate ambiguity, 4: Slight ambiguity, 5: No ambiguity at all] Please rate the degree of ambiguity you perceive in the true relevance labels on a scale from 1 to 5.
- **To what degree do you think the process of answering multiple choice clinical QAs correlates with real-world day-to-day medical practice?** [1: Not at all aligned, 2: Slightly aligned, 3: Moderately aligned, 4: Strongly aligned, 5: Very closely aligned] To what degree do you think the process of answering multiple choice clinical QAs correlates with real-world day-to-day medical practice?
- **Approximately what percentage of questions did you skip?** [Short answer: 0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100] Approximately what percentage of questions did you skip?

6.2.1 Physician Survey Analysis

All 7 physicians have never seen the QAs. Their average of accuracy is 60%, which is aligned with the final accuracy performance of 62.4% (SD = 0.5). The average of the physicians' assumptions on best LLM's performance is 70%, and the best LLM GPT-5 has the performance of 86.4%. Most physicians are confidence in the sentence-level relevance scores estimation, with the average of the ambiguity in relevance labels is 3 (Moderately Confident), SD = 1.07. Most physicians also indicated that answering multiple choice clinical QAs slightly correlates with real-world day-to-day medical practice. In average, the physician labelers skipped 26.4% of QAs (SD = 21.2) because the QAs are out of their expertise or other reasons.

6.2.2 Trainee Survey Analysis

In the prestudy survey all 52 physician trainee labelers provided demographic and background information. The cohort was predominantly male (67.3%), with female annotators accounting for 30.8%. Most participants (85.7%) were in the senior years of their medical training, and 72.2% had already passed the USMLE Step 1.

Rarecase resources were well known to the group: 52.8% reported familiarity with collections such as the JAMA Clinical Challenge, NEJM Image Challenge, and NEJM Resident 360, and 41.7% stated that they consult these materials regularly. Exposure to clinical large language models was also high: 91.7% had used or observed LLMs in practice, whether for answering clinical questions or integrating decisionsupport functions. On average, respondents estimated that 49.0% (SD = 21.8) of existing LLMs are sufficiently mature for deployment in clinical settings.

We received 29 post-study surveys from the 52 participating labelers. On average, only 8.6% (SD = 0.2) of the questions they labeled had been encountered verbatim before the study. Labelers expected LLMs to outperform them in the QA answering performance, predicting 79.7% accuracy for the models (SD = 0.1) versus 63.5% for themselves (SD= 0.2).

When reflecting on their sentencelevel relevance judgments, a majority of 72.4% described themselves as moderately confident, an 58.6% characterized the task as exhibiting moderate ambiguity. Regarding how well the multiple choice QA format matches real-world clinical practice, 44. 8% viewed the multiple choice QA format as 'moderately aligned' with clinical practice, while 37. 9% considered it 'slightly aligned'. As labelers are allowed to skip questions if they do not know the answer, an average of 20.9% (12.9) of questions are skipped, which demonstrates the difficulties of the QAs. These qualitative feedback suggests that relevance labels represent consensusbased approximations rather than definitive ground truth, and that QA benchmarks should be complemented by richer, practicegrounded evaluations in future work.

Both the physicians' pre-study and post-study survey results are available in Table 9 and Table 10; the physician trainees' pre-study and post-study survey results are available in Table 11 and Table ??.

6.3 Physician and Physician Trainee Labeler Information

Table 9: **Pre-Survey Demographics and Educational Background of Physician Participants, Including Self-Reported Familiarity with Clinical Challenges and LLMs in Healthcare.** “N/A” denotes that the labeler chose not to disclose this information. The complete list of pre-survey questions is available in the supplementary material.

ID	Age	Gender	Year of Postgraduate Year (PGY)	USMLE Step 3 Passed?	Medical school	Familiarity with clinical challenges?	How regularly follow challenges?	Familiarity with MedBullets	How regularly follow MB?	Familiarity with LLMs in healthcare	LLM Clinical deployment readiness
01	41	Male	15	Yes	University of Rochester	Some familiarity	Some familiarity	Some familiarity	Occasionally	Some familiarity	Ready for limited, low-risk tasks
02	53	Female	24	Yes	Wright State University School of Medicine	Not familiar	Not familiar	Some familiarity	Not at all	High familiarity	Ready for limited, low-risk tasks
04	44	Female	17	Yes	Brown Medical School	Some familiarity	High familiarity	Not familiar	Not at all	High familiarity	Ready for clinician-in-the-loop tasks
05	42	Female	17	Yes	RWJMS New Brunswick	Some familiarity	Some familiarity	Some familiarity	Not at all	High familiarity	Ready for clinician-in-the-loop tasks
06	41	Male	17	UK Based	Aagazig University Faculty of Medicine	Not familiar	Not familiar	Not familiar	Occasionally	High familiarity	Ready for clinician-in-the-loop tasks
07	45	Male	Attending	Yes	Universidad Anahuac (SP)	Some familiarity	Some familiarity	Not familiar	Not at all	High familiarity	Ready for clinician-in-the-loop tasks

Table 10: **Post-Survey of Physician Participants.**

Origin_ID	Percentage seen verbatim before? (%)	Estimated accuracy percentage? (%)	Assumption on Best LLM's accuracy? (%)	Confidence in relevance scores?	Ambiguity in relevance labels?	Correlation with real-world practice?	Percentage of questions skipped? (%)
01	0	40	70	Moderately Confident	2: High ambiguity	Slightly aligned	61 - 70
02	0	70	80	Moderately Confident	3: Moderate ambiguity	Slightly aligned	21 - 30
03	0	90	80	Extremely Confident	5: No ambiguity at all	Not at all aligned	0 -10
04	0	60	90	Moderately Confident	3: Moderate ambiguity	Strongly aligned	31 - 40
05	0	30	40	Slightly confident	2: High ambiguity	Not at all aligned	11 - 20
06	0	70	90	Moderately Confident	4: Slight ambiguity	Moderately aligned	31 - 40
07	0	60	40	Moderately Confident	2: High ambiguity	Slightly aligned	0 -10

Table 11: **Pre-Survey Demographics and Educational Background of Medical Student Participants, Including Self-Reported Familiarity with Clinical Challenges and LLMs in Healthcare.** Empty cells denotes that the labeler chose not to disclose this information. The complete list of pre-survey questions is available in the supplementary material.

ID	Age	Gender	Year of medical school	USMLE Step 1 passed?	Medical school	Familiarity with clinical challenges	How regularly follow challenges	Familiarity with MedBullets	How regularly follow MB	Familiarity with LLMs	LLM deployment readiness
1	26	Male	M3	No	Rush Medical College	Some familiarity	Occasionally	Some familiarity	Occasionally	High familiarity	25
2	22	Male	M1	No	Dartmouth Geisel School of Medicine	Some familiarity	Not at all	High familiarity	Occasionally	High familiarity	80
3	27	Female	M3	Yes	UNC Chapel Hill SOM	Not familiar	N/A	Some familiarity	Not at all	Some familiarity	30
4	26	Female	M3	No	Tufts University School of Medicine	Not familiar	N/A	Not familiar	N/A	Some familiarity	70
5		Male	M4	Yes	Dell Medical School	Not familiar	N/A	Some familiarity	Not at all	Not familiar	20
6	25	Male	M3	Yes	Warren Alpert Medical School of Brown University	Some familiarity	Occasionally	High familiarity	Not at all	Some familiarity	45
7	26	Female	M4	Yes	George Washington University SOM	High familiarity	Occasionally	High familiarity	Occasionally	High familiarity	90
8	28	Male	G3 (MD/PhD)	Yes	UC San Diego	Some familiarity	Not at all	Not familiar	N/A	High familiarity	30
9	29	Male	M4	Yes	Emory University	Some familiarity	Occasionally	Not familiar	Not at all	High familiarity	60
10	25	Male	M3	Yes	Indiana University School of Medicine	Some familiarity	N/A	Some familiarity	Not at all	High familiarity	58
11	26	Female	M3	Yes	Tufts University School of Medicine	Not familiar	Not at all	Not familiar	Not at all	Some familiarity	40
12	24	Male	M3	Yes	Columbia VP&S	Not familiar	N/A	Not familiar	N/A	High familiarity	20
13		Male	M2	Yes	NYU Grossman School of Medicine	Some familiarity	Not at all	Not familiar	Not at all	Some familiarity	70
14	27	gender queer	M3	Yes	KPSOM	Some familiarity	Not at all	Some familiarity	Not at all	Not familiar	NA
15		Male	M2	Yes	Harvard	High familiarity	Occasionally	High familiarity	Occasionally	High familiarity	25
16	26	Male	M4	Yes	Dell Medical school	Some familiarity	Not at all	Not familiar	Not at all	Some familiarity	30
17	32	Male	M4	Yes	Northwestern	Not familiar	Not at all	Some familiarity	Not at all	Some familiarity	5
18	25	Male	M1	No	University of Texas Medical Branch	Some familiarity	Occasionally	Not familiar	Not at all	Some familiarity	50
19	28	Female	M3	Yes	KPSOM	Some familiarity	Occasionally	High familiarity	Occasionally	High familiarity	40
20	26	Male	M4	Yes	UNC School of Medicine	Some familiarity	Occasionally	High familiarity	Occasionally	High familiarity	75
21	27	Female	M4	Yes	Alabama College of Osteopathic Medicine	Not familiar	N/A	Not familiar	N/A	Some familiarity	50
22	23	Male	M1	No	University of Maryland School of medicine	Some familiarity	Occasionally	Some familiarity	Not at all	Some familiarity	45
23	33	Female	M4	Yes	Touro College of Osteopathic Medicine Middletown NY	Not familiar	N/A	Not familiar	N/A	Some familiarity	70
24		Female	M3	No	Case Western	Some familiarity	Not at all	High familiarity	Not at all	High familiarity	30%
25		Male	M4	Yes	Harvard	High familiarity	Occasionally	Not familiar	Not at all	High familiarity	60
26	25	Male	M3	Yes	UNC School of Medicine	Not familiar	N/A	Not familiar	N/A	High familiarity	80
27	23	Male	M1	No	UC Irvine	Not familiar	Not at all	Not familiar	Not at all	High familiarity	50
28	25	Male	M3	No	WUSM	Not familiar	Not at all	Some familiarity	Not at all	High familiarity	70
29	25	Male	M3	Yes	Indiana University School of Medicine	Some familiarity	Not at all	Some familiarity	Not at all	High familiarity	20
30	27	Female	M3	Yes	UNC	Not familiar	Not at all	Some familiarity	Not at all	Some familiarity	35
31	26	Female	M3	No	Tufts University School of Medicine	Not familiar	Not at all	Not familiar	N/A	Not familiar	70
32	22	Male	M1	No	Dartmouth Geisel SOM	Some familiarity	Occasionally	Some familiarity	Occasionally	Some familiarity	75
33	28	Male	M4	Yes	Alabama College of Osteopathic Medicine	Some familiarity	Occasionally	Not familiar	Not at all	Some familiarity	60
34	25	Male	M4	Yes	University of Toledo	Some familiarity	Not at all	High familiarity	Not at all	Some familiarity	25%
35	26	Male	M4	Yes	UNC Chapel Hill	Not familiar	Not at all	Not familiar	Not at all	Some familiarity	30
36	26	Male	M4	Yes	UNC School of Medicine	Some familiarity	Occasionally	Some familiarity	Occasionally	High familiarity	40

(Continued on next page)

(Continued from previous page)

ID	Age	Gender	Year of medical school	USMLE Step 1 passed?	Medical school	Familiarity with clinical challenges	How regularly follow challenges	Familiarity with MedBullets	How regularly follow MB	Familiarity with LLMs	LLM deployment readiness
37	26	Female	M4	Yes	Medical College of Georgia	Not familiar	N/A	Some familiarity	Not at all	Some familiarity	45
38	25	Male	M4	Yes	University of Toledo	Some familiarity	Occasionally	Some familiarity	Occasionally	High familiarity	80
39	28	Female	M4	Yes	Dartmouth	Not familiar	N/A	Some familiarity	Not at all	High familiarity	80
40	29	Male	M4	Yes	Emory	Some familiarity	Occasionally	Not familiar	Not at all	High familiarity	60
41	27	Male	M3	Yes	University of Cincinnati	Not familiar	Not at all	Some familiarity	Occasionally	High familiarity	75
42	25	Male	M4	Yes	University of Toledo	Some familiarity	Not at all	High familiarity	Not at all	Some familiarity	25%
43	28	Male	Graduate School	Yes	UC San Diego	Not familiar	Not at all	Not familiar	Not at all	High familiarity	50
44	28	Female	M4	Yes	Geisel School of Medicine at Dartmouth	Some familiarity	Not at all	Not familiar	N/A	Not familiar	10
45	33	Female	M4	Yes	TouroCOM NY	Not familiar	N/A	Not familiar	N/A	Not familiar	50
46	30	Male	M4	Yes	Dell Medical School	Not familiar	Not at all	Not familiar	Not at all	Some familiarity	50
47	25	Male	M3	Yes	Columbia VP&S	Not familiar	N/A	Not familiar	N/A	Some familiarity	10
48	26	Male	M4	Yes	Dell Medical School	Not familiar	N/A	Not familiar	N/A	High familiarity	40
49	25	Male	M1	No	St Georges University School of Medicine	Some familiarity	Not at all	Not familiar	Not at all	High familiarity	75
50	28	Female	M4	Yes	Geisel School of Medicine	Not familiar	N/A	Some familiarity	Not at all	Some familiarity	55
51	29	Male	M4	Yes	UCLA	High familiarity	Occasionally	Some familiarity	Not at all	High familiarity	70
52	24	Female	M1	No	kansas city university	Not familiar	Not at all	Not familiar	Not at all	Some familiarity	50

Table 12: **Round 2 Physician-Trainee Highly Relevant Labeled Sentences: Medical Student Participants Demographics.** Empty cells denote that the labeler chose not to disclose this information.

ID	Age	M/F	Year Med Sch	USMLE Passed	Prev Qs	Prev Ans	Medical school	Clinical Challenge Famil	Challenge Freq	MedBullets Famil	MedBullets Freq	LLM Famil	LLM Deploy Ready
63	30	Female	M4	Yes	N/A		N/A	Not familiar	Occasionally	High familiarity	Not at all	Some familiarity	Ready for limited, low-risk tasks
64	27	Male	M4	Yes	0	N/A	Indiana University School of Medicine	Not familiar	Not at all	Not familiar	Not at all	Some familiarity	Ready for clinician-in-the-loop tasks
65	25	Male	M3	Yes	0	N/A	Case Western Reserve SOM	Not familiar	Not at all	Not familiar	Not at all	Some familiarity	Ready for limited, low-risk tasks
66	29	Male	M4	Yes	0	N/A	Brown	Some familiarity	Not at all	Some familiarity	Occasionally	Some familiarity	Ready for clinician-in-the-loop tasks
67	26	Male	M3	Yes	0	N/A	Loma Linda University	High familiarity	Occasionally	Not familiar	Not at all	Some familiarity	Ready for clinician-in-the-loop tasks
68	28	Male	M4	Yes	0	N/A	UCLA	Not familiar	Not at all	Some familiarity	Not at all	Some familiarity	Ready for limited, low-risk tasks
69	27	Female	M4	Yes	0	N/A	Texas Tech University Health Sciences Center SOM	Some familiarity	Not at all	Not familiar	Not at all	Some familiarity	Ready for limited, low-risk tasks
70	27	Female	M4	Yes	0	N/A	CUNY School of Medicine	Not familiar	Not at all	Some familiarity	Occasionally	Some familiarity	Ready for limited, low-risk tasks
71	N/A	Female	M4	Yes	5	N/A		Not familiar	Not at all	Not familiar	Not at all	Some familiarity	Ready for limited, low-risk tasks
72	24	Female	M1	N/A		0	University of Texas Rio Grande Valley School of Medicine	Some familiarity	Not at all	Not familiar	Not at all	Some familiarity	Ready for clinician-in-the-loop tasks
73	31	Female	M3	Yes	0	N/A	Trinity School of Medicine	Not familiar	Not at all	Some familiarity	Not at all	Some familiarity	Ready for clinician-in-the-loop tasks

Table 13: **Pre-Survey Demographics and Educational Background of Round 3 (GPT-5) Medical Student Participants, Including Self-Reported Familiarity with Clinical Challenges and LLMs in Healthcare.** Empty cells denote that the labeler chose not to disclose this information. The complete list of pre-survey questions is available in the supplementary material.

ID	Age	Gender	Year of medical school	USMLE Step 1 passed?	Medical school	Familiarity with clinical challenges	How regularly follow challenges	Familiarity with MedBullets	How regularly follow MB	Familiarity with LLMs	LLM deployment readiness
74	N/A	Male	M3	N/A	Brown	Not familiar	Not at all	Some familiarity	Not at all	Some familiarity	Ready for limited, low-risk tasks
75	29	Female	M4	Yes	UNC School of Medicine	Some familiarity	Occasionally	High familiarity	Occasionally	High familiarity	Ready for clinician-in-the-loop tasks
76	27	Male	N/A	Yes	UNC SOM graduate	Some familiarity	N/A	High familiarity	Not at all	High familiarity	Ready for limited, low-risk tasks

Table 14: **Pre-Survey Demographics and Educational Background of Round 4 (Qwen-72B) Medical Student Participants, Including Self-Reported Familiarity with Clinical Challenges and LLMs in Healthcare.** Empty cells denote that the labeler chose not to disclose this information. The complete list of pre-survey questions is available in the supplementary material.

ID	Age	Gender	Year of medical school	USMLE Step 1 passed?	Medical school	Familiarity with clinical challenges	How regularly follow challenges	Familiarity with MedBullets	How regularly follow MB	Familiarity with LLMs	LLM deployment readiness
77	28	Male	M4	Yes	Dartmouth College	Some familiarity	Occasionally	High familiarity	Occasionally	Some familiarity	Ready for limited, low-risk tasks
78	26	Female	M4	Yes	Geisel Dartmouth School of Med	Not familiar	Not at all	Some familiarity	Not at all	Some familiarity	Ready for limited, low-risk tasks
79	27	Female	M4	Yes	Temple Lewis Katz School of Medicine	Not familiar	Not at all	Some familiarity	Not at all	Some familiarity	Ready for limited, low-risk tasks
80	23	Male	M2	No	N/A	Not familiar	N/A	High familiarity	N/A	Some familiarity	Ready for limited, low-risk tasks

Table 15: **Post-Survey Exit Feedback from All Study Participants.** Column Appendix ID corresponds to labeler ID in Tables 10-14. The complete list of pre-survey questions is available in the supplementary material.

Appendix ID	Seen verbatim (%)	Estimated accuracy (%)	Best LLM accuracy (%)	Confidence	Ambiguity	Real-world	Skipped (%)
1	0	70	80	Moderately Confident	3: Moderate ambiguity	Slightly aligned	21 - 30
2	0	60	90	Moderately Confident	3: Moderate ambiguity	Strongly aligned	31 - 40
3	0	40	70	Moderately Confident	2: High ambiguity	Slightly aligned	61 - 70
4	0	30	40	Slightly confident	2: High ambiguity	Not at all aligned	11 - 20
5	0	90	80	Extremely Confident	5: No ambiguity at all	Not at all aligned	0 - 10
6	0	70	90	Moderately Confident	4: Slight ambiguity	Moderately aligned	31 - 40
7	0	60	40	Moderately Confident	2: High ambiguity	Slightly aligned	0 - 10
8	80	80	100	Moderately Confident	3: Moderate ambiguity	Not at all aligned	0 - 10
9	0	30	70	Moderately Confident	4: Slight ambiguity	Slightly aligned	0 - 10
53	50	70	90	N/A	4: Slight ambiguity	Strongly aligned	11 - 20
54	0	50	80	Moderately Confident	4: Slight ambiguity	Slightly aligned	0 - 10
55	0	30	70	Moderately Confident	3: Moderate ambiguity	Slightly aligned	0 - 10
56	0	70	90	Very Confident	3: Moderate ambiguity	Moderately aligned	0 - 10
57	0	70	70	Slightly confident	3: Moderate ambiguity	Slightly aligned	0 - 10
58	0	60	90	Moderately Confident	4: Slight ambiguity	Slightly aligned	0 - 10
59	0	50	90	Slightly confident	3: Moderate ambiguity	Strongly aligned	0 - 10
60	0	30	70	Not Confident	3: Moderate ambiguity	Slightly aligned	0 - 10
61	10	70	90	Moderately Confident	4: Slight ambiguity	Moderately aligned	0 - 10
62	0	60	80	Moderately Confident	3: Moderate ambiguity	Slightly aligned	0 - 10
63	0	20	80	Slightly confident	3: Moderate ambiguity	Strongly aligned	0 - 10
64	0	60	90	Slightly confident	4: Slight ambiguity	Moderately aligned	0 - 10
65	60	60	80	Moderately Confident	3: Moderate ambiguity	Moderately aligned	0 - 10
66	10	80	80	Moderately Confident	3: Moderate ambiguity	Slightly aligned	11 - 20
69	0	60	70	Moderately Confident	3: Moderate ambiguity	Slightly aligned	0 - 10
70	10	10	20	Not Confident	2: High ambiguity	Not at all aligned	0 - 10
71	0	20	80	Slightly confident	2: High ambiguity	Not at all aligned	0 - 10
72	10	60	90	Moderately Confident	3: Moderate ambiguity	Slightly aligned	0 - 10
73	30	80	80	Slightly confident	3: Moderate ambiguity	Slightly aligned	0 - 10
74	0	70	60	Slightly confident	2: High ambiguity	Slightly aligned	0 - 10
75	0	70	90	Very Confident	3: Moderate ambiguity	Slightly aligned	21 - 30
76	20	70	90	Moderately Confident	3: Moderate ambiguity	Slightly aligned	0 - 10
77	20	30	80	Moderately Confident	3: Moderate ambiguity	Moderately aligned	0 - 10
79	0	30	90	Not Confident	2: High ambiguity	Slightly aligned	0 - 10
80	10	10	10	Slightly confident	2: High ambiguity	Moderately aligned	0 - 10